biprism and the eyepiece. For a fixed position of the eyepiece there will be two positions of the lens (shown as L_1 and L_2 in Fig. 14.19) where the images of S_1 and S_2 can be seen at the eyepiece.* Let d_1 be the distance between the two images when the lens is at the position L_1 (at a distance b_1 from the eyepiece). Let d_2 and b_2 be the corresponding distances when the lens is at L_2. Then, it can easily be shown that

$$d = \sqrt{d_1 d_2}$$

and

$$D = b_1 + b_2$$

Typically for $d \approx 0.01$ cm, $\lambda \approx 6 \times 10^{-5}$ cm, $D \approx 50$ cm, $\beta \approx 0.3$ cm.

In the above arrangement we have considered a slit instead of a point source. Since each pair of points S_1 and S_2 produce (approximately) straight line fringes, the slit will also produce straight line fringes of increased intensity.

14.9 INTERFERENCE WITH WHITE LIGHT

LO 7

We will now discuss the interference pattern when the slit is illuminated by white light. The wavelengths corresponding to the violet and red end of the spectrum are about 4×10^{-5} cm and 7×10^{-5} cm, respectively. Clearly, the central fringe produced at the point O (Fig. 14.19) will be white because all wavelengths will constructively interfere here. Now, slightly below (or above) the point O the fringes will become colored. For example, if the point P is such that

$$S_2P \sim S_1P = 2 \times 10^{-5} \text{ cm} \left(= \frac{\lambda_{\text{violet}}}{2} \right)$$

then complete destructive interference will occur only for the violet color. Partial destructive interference will occur for other wavelengths. Consequently, we will have a line devoid of the violet color and will appear reddish. The point Q which satisfies

$$S_2Q \sim S_1Q = 3.5 \times 10^{-5} \text{ cm} \left(= \frac{\lambda_{\text{red}}}{2} \right)$$

will be devoid of the red color. It will correspond to almost constructive interference for the violet color. No other wavelength (in the visible region) will neither constructively nor destructively interfere. Thus following the white central fringe we will have colored fringes; when the path difference is about 2×10^{-5} cm the fringe will be red in color, then the color will gradually change to violet. The colored fringes will soon disappear because at points far away from O there will be so many wavelengths (in the visible region) which will constructively interfere that we will observe uniform white illumination. For example, at a point R, such that $S_2R \sim S_1R = 30 \times 10^{-5}$ cm, wavelengths corresponding to $30 \times 10^{-5}/n$ ($n = 1, 2, \ldots$) will constructively interfere. In the visible region, these wavelengths will be 7.5×10^{-5} cm (red), 6×10^{-5} cm (yellow), 5×10^{-5} cm (greenish yellow) and 4.3×10^{-5} cm (violet). Further, wavelengths corresponding to $30 \times 10^{-5}/\left(n + \frac{1}{2}\right)$ will destructively interfere; thus, in the visible region, the wavelengths 6.67×10^{-5} cm (orange), 5.5×10^{-5} cm (yellow), 4.6×10^{-5} cm (indigo) and 4.0×10^{-5} cm (violet) will be absent. The color of such light, as seen by the unaided eye, will be white. Thus, with white light one gets a white central fringe at the point of zero path difference along with a few colored fringes on both the sides, the color soon fading off to white. While using a white light source, if we put a red (or green) filter in front of our eye, we will see the interference pattern corresponding to the red (or green) light.

In the usual interference pattern with a nearly monochromatic source (like a sodium lamp) a large number of interference fringes are obtained and it is extremely difficult to determine the position of the central fringe. In many interference experiments, it is necessary to determine the position of the central fringe and, as has been discussed above, this can easily be done by using white light as a source.

As discussed above, when we observe interference pattern using a white light source, we will see only few colored fringes. However, if we put a red filter in front of our eye, the fringe pattern (corresponding to the red color) will suddenly appear. If we replace the red filter by a green filter in front of our eye, the fringe pattern corresponding to the green color will appear.

14.10 DISPLACEMENT OF FRINGES

LO 8

We will now discuss the change in the interference pattern produced by introducing a thin transparent plate in the path of one of the two interference beams as shown in Fig. 14.20. Let t be the thickness of the plate and let n be its refractive

* This method is similar to the displacement method for the determination of the focal length of a convex lens.

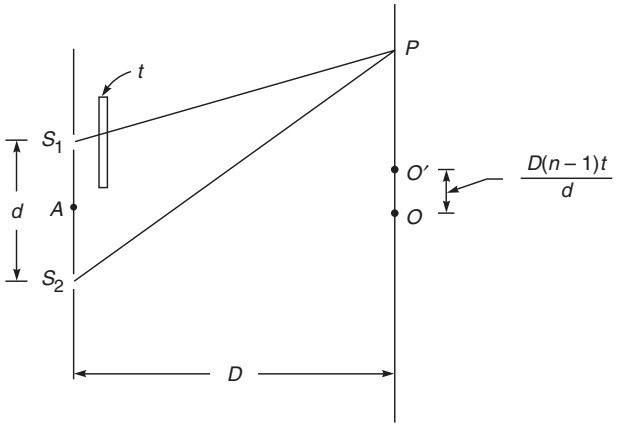


Fig. 14.20 If a thin transparent sheet (of thickness t) is introduced in one of the beams, the fringe pattern gets shifted by a distance $(n - 1)\, t\, D/d$.

index. It is easily seen from the figure that light reaching the point P from S_1 has to traverse a distance t in the plate and a distance $S_1P - t$ in air. Thus the time required for the light to reach from S_1 to the point P is given by

$$\frac{S_1P - t}{c} + \frac{t}{v} = \frac{1}{c}[S_1P - t + nt]$$

$$= \frac{1}{c}\,[S_1P + (n-1)t] \qquad \textbf{(14.35)}$$

where $v\ \left(=\frac{c}{n}\right)$ represents the speed of light in the plate. Equation (14.35) shows that by introducing the thin plate the effective optical path increases by $(n - 1)t$. Thus, when the thin plate is introduced, the central fringe (which corresponds to equal optical path from S_1 and S_2) is formed at the point O' where

$$S_1O' + (n-1)t = S_2O'$$

Since [see Eq. (14.19)]

$$S_2O' - S_1O' \approx \frac{d}{D}\ OO'$$

therefore,

$$(n-1)t = \frac{d}{D}\ OO' \qquad \textbf{(14.36)}$$

Thus the fringe pattern gets shifted by a distance Δ which is given by the following equation:

$$\Delta = \frac{D(n-1)\,t}{d} \qquad \textbf{(14.37)}$$

The above principle enables us to determine the thickness of extremely thin transparent sheets (like that of mica) by measuring the displacement of the central fringe. Further, if white light is used as a source, the displacement of the central fringe is easy to measure.

Example 14.10 In a double-slit interference arrangement, one of the slits is covered by a thin mica sheet whose refractive index is 1.58. The distances S_1S_2 and AO (see Fig. 14.20) are 0.1 cm and 50 cm, respectively. Due to the introduction of the mica sheet the central fringe gets shifted by 0.2 cm. Determine the thickness of the mica sheet.

Solution: $\Delta = 0.2$ cm; $d = 0.1$ cm; $D = 50$ cm

Hence,

$$t = \frac{d\Delta}{D(n-1)} = \frac{0.1 \times 0.2}{50 \times 0.58}$$

$$\approx 6.7 \times 10^{-4} \text{ cm}$$

Example 14.11 In an experimental arrangement similar to the one discussed in the above example, one finds that by introducing the mica sheet the central fringe occupies the position that was originally occupied by the eleventh bright fringe. If the source of light is a sodium lamp ($\lambda = 5893$ Å) determine the thickness of the mica sheet.

Solution: The point O' (see Fig. 14.20) corresponds to the eleventh bright fringe, thus,

$$S_2O' - S_1O' = 11\lambda = (n-1)t = 0.58t$$

14.11 THE LLOYD'S MIRROR ARRANGEMENT LO 9

In this arrangement, light from a slit S_1 is allowed to fall on a plane mirror at grazing incidence (see Fig. 14.21). The light directly coming from the slit S_1 interferes with the light reflected from the mirror forming an interference pattern in the region BC of the screen. One may thus consider the slit S_1 and its virtual image S_2 to form two coherent sources which produce the interference pattern. It should be noted that at grazing incidence one really need not have a mirror; even a dielectric surface has very high reflectivity (see Chapter 23).

As can be seen from Fig. 14.21, the central fringe cannot be observed on the screen unless the latter is moved to the position $L'_1L'_2$, where it touches the end of the reflector. Alternatively, one may introduce a thin mica sheet in the path of the direct beam so that the central fringe appears in the region BC. (This is discussed in detail in Problem 14.2) Indeed, if the central fringe is observed with white light, it is found to be dark. This implies that the reflected beam under-

goes a sudden phase change of π on reflection. Consequently, when the point P on the screen is such that

$$S_2P - S_1P = n\lambda, n = 0, 1, 2, 3,\ldots$$

we will get minima (i.e., destructive interference). On the other hand, if

$$S_2P - S_1P = \left(n + \frac{1}{2}\right)\lambda$$

we will get maxima.

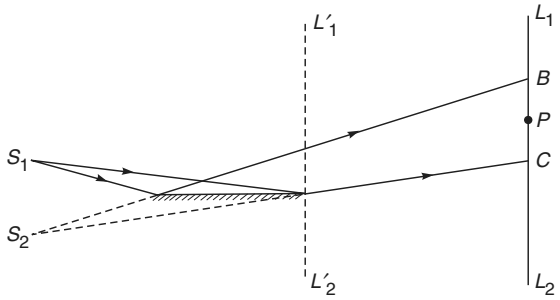


Fig. 14.21 The Lloyd's mirror arrangement.

In the next section, using the principle of optical reversibility, we will show that if there is an abrupt phase change of π when light gets reflected by a denser medium, then no such abrupt phase change occurs when reflection takes place at a rarer medium.

14.12 PHASE CHANGE ON REFLECTION

We will now investigate the reflection of light at an interface between two media using the principle of optical reversibility. According to this principle, in the absence of any absorption, a light ray that is reflected or refracted will retrace its original path if its direction is reversed.*

Consider a light ray incident on an interface of two media of refractive indices n_1 and n_2 as shown in Fig. 14.22(a). Let the amplitude reflection and transmission coefficients be r_1 and t_1, respectively. Thus, if the amplitude of the incident ray is a, then the amplitudes of the reflected and refracted rays would be ar_1 and at_1, respectively.

We now reverse the rays and we consider a ray of amplitude at_1 incident on medium 1 and a ray of amplitude ar_1 incident on medium 2 as shown in Fig. 14.22(b). The ray of amplitude at_1 will give rise to a reflected ray of amplitude at_1r_2 and a transmitted ray of amplitude at_1t_2 where r_2 and t_2 are the amplitude reflection and transmission coefficients when a ray is incident from medium 2 on medium 1. Similarly, the ray of amplitude ar_1 will give rise to a ray of amplitude ar_1^2 and a refracted ray of amplitude ar_1t_1. According to the principle of optical reversibility the two rays of amplitudes ar_1^2 and at_1t_2 must combine to give the incident ray of Fig. 14.22(a); thus,

$$ar_1^2 + at_1t_2 = a$$

or

$$t_1t_2 = 1 - r_1^2 \qquad \textbf{(14.38)}$$

Further, the two rays of amplitudes at_1r_2 and ar_1t_1 must cancel each other, i.e.,

$$at_1r_2 + ar_1t_1 = 0$$

or

$$r_2 = -r_1 \qquad \textbf{(14.39)}$$

Since we know from the Lloyd's mirror experiment that an abrupt phase change of π occurs when light gets reflected by a denser medium, we may infer from Eq. (14.39) that no such abrupt phase change occurs when light gets reflected by a rarer medium. This is indeed borne out by experiments. Equations (14.38) and (14.39) are known as **Stokes' relations**.

In Chapter 24, we will calculate the amplitude reflection and transmission coefficients for plane waves incident on a dielectric and also on a conductor. It will be shown that the coefficients satisfy Stokes' relations; the phase change on reflection will also be discussed there.

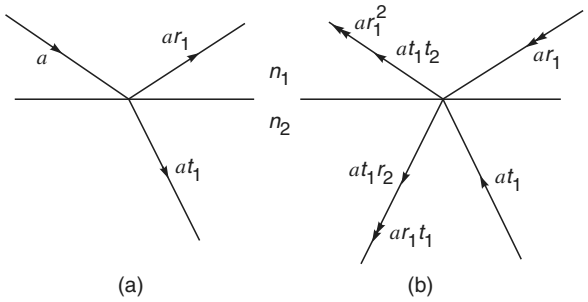


Fig. 14.22 (a) A ray traveling in a medium of refractive index n_1 incident on a medium of refractive index n_2. (b) Rays of amplitude ar_1 and at_1 incident on a medium of refractive index n_1.

* This principle is consequence of time reversal invariance according to which processes can run either way in time; for more details see Refs. 14.3 and 14.8.

Summary

- In 1801, Thomas Young devised an ingenious but simple method to lock the phase relationship between two sources of light. The trick lies in the division of a single wavefront into two; these two split wavefronts act as if they emanated from two sources having a fixed phase relationship and, therefore, when these two waves were allowed to interfere, a stationary interference pattern is obtained.
- For two coherent point sources, almost straight-line interference fringes are formed on some planes and by measuring the fringe width (which represents the distance between two consecutive fringes) one can calculate the wavelength.
- On a plane which is normal to the line joining the two coherent point sources, the fringe pattern is circular.
- In the Young's double-slit interference pattern, if we use a white light source, one gets a white central fringe at the point of zero path difference along with a few colored fringes on both the sides, the color soon fading off to white. If we now introduce a very thin slice of transparent material (like mica) in the path of one of the interfering beams, the fringes get displaced and by measuring the displacement of fringes, one can calculate the thickness of the mica sheet.

Problems

14.1 In the Young's double-hole experiment (see Fig. 14.6), the distance between the two holes is 0.5 mm, $\lambda = 5 \times 10^{-5}$ cm and D = 50 cm. What will be the fringe width?

14.2 Figure 14.23 represents the layout of Lloyd's mirror experiment. S is a point source emitting waves of frequency 6×10^{14} s^{-1}. A and B represent the two ends of a mirror placed horizontally and LOM represents the screen. The distances SP, PA, AB and BO are 1 mm, 5 cm, 5 cm and 190 cm, respectively. (a) Determine the position of the region where the fringes will be visible and calculate the number of fringes. (b) Calculate the thickness of a mica sheet (n = 1.5) which should be introduced in the path of the direct ray so that the lowest fringe becomes the central fringe. The velocity of light is 3×10^{10} cm/s.

[**Ans:** (a) 2 cm, 40 fringes, (b) 38 μm]

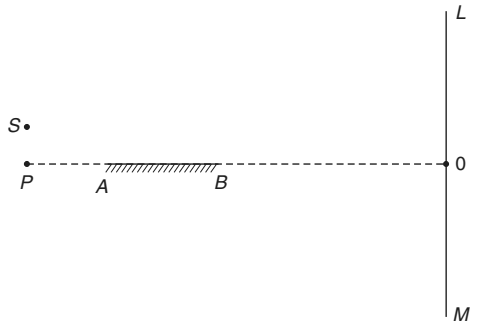


Fig. 14.23 Figure for Problem 14.2.

14.3 **(a)** In the Fresnel's biprism arrangement, show that $d = 2(n - 1)\, a\alpha$ where a represents the distance from the source to the base of the prism (see Fig. 14.19), α is the angle of the biprism and n the refractive index of the material of the biprism.

(b) In a typical biprism arrangement b/a = 20, and for sodium light ($\lambda \simeq$ 5893 Å) one obtains a fringe width of 0.1 cm; here b is the distance between the biprism and the screen. Assuming n = 1.5, calculate the angle α.

[**Ans:** $\simeq$ 0.71°]

14.4 In the Young's double-hole experiment a thin mica sheet (n = 1.5) is introduced in the path of one of the beams. If the central fringe gets shifted by 0.2 cm, calculate the thickness of the mica sheet. Assume d = 0.1 cm, and D = 50 cm.

14.5 In order to determine the distance between the slits in the Fresnel biprism experiment, one puts a convex lens in between the biprism and the eyepiece. Show that if $D > 4f$ one will obtain two positions of the lens where the image of the slits will be formed at the eyepiece; here f is the focal length of the convex lens and D is the distance between the slit and the eyepiece. If d_1 and d_2 are the distances between the images (of the slits) as measured by the eyepiece, then show that $d = \sqrt{d_1 d_2}$. What would happen if $D < 4f$?

14.6 In the Young's double hole experiment, interference fringes are formed using sodium light which predominantly comprises of two wavelengths (5890 Å and 5896 Å). Obtain the regions on the screen where the fringe pattern will disappear. You may assume d = 0.5 mm and D = 100 cm.

14.7 If one carries out the Young's double-hole interference experiment using microwaves of wavelength 3 cm, discuss the nature of the fringe pattern if d = 0.1 cm, 1 cm, and 4 cm. You may assume D = 100 cm. Can you use Eq. (14.21) for the fringe width?

14.8 In the Fresnel's two-mirror arrangement (see Fig. 14.18) show that the points S, S_1, and S_2 lie on a circle and $S_1S_2 = 2b\theta$ where $b = MS$ and θ is the angle between the mirrors.

14.9 In the double-hole experiment using white light, consider two points on the screen, one corresponding to a path difference of 5000 Å and the other corresponding to a path difference of 40000 Å. Find the wavelengths (in the visible region) which correspond to constructive and destructive interference. What will be the color of these points?

14.10 **(a)** Consider a plane which is normal to the line joining two point coherent sources S_1 and S_2 as shown in Fig. 14.14. If $S_1P - S_2P = \Delta$, then show that

$$y = \frac{1}{2\Delta}(d^2 - \Delta^2)^{\frac{1}{2}}[4D^2 + 4Dd + (d^2 - \Delta^2)]^{\frac{1}{2}}$$

$$\approx \frac{D}{\Delta}\sqrt{(d - \Delta)(d + \Delta)}$$

where the last expression is valid for $D >> d$.

(b) For λ = 0.5 μm, d = 0.4 mm and D = 20 cm; $S_1O - S_2O = 800\,\lambda$. Calculate the value of $S_1P - S_2P$ for the point P to be first dark ring and first bright ring.

[**Ans:** 0.39975 mm, 0.3995 mm]

14.11 In continuation of the above problem calculate the radii of the first two dark rings for

(a) D = 20 cm and **(b)** D = 10 cm.

[**Ans:** (a) ≈ 0.71 cm and 1.22 cm]

14.12 In continuation of Problem 14.10 assume that d = 0.5 mm, $\lambda = 5 \times 10^{-5}$ cm and D = 100 cm. Thus the central (bright) spot will correspond to n = 1000. Calculate the radii of the first, second and third bright rings which will correspond to n = 999, 998 and n = 997, respectively.

14.13 Using the expressions for the amplitude reflection and transmission coefficients (derived in Chapter 24), show that they satisfy Stokes' relations.

14.14 Assume a plane wave incident normally on a plane containing two holes separated by a distance d. If we place a convex lens behind the slits, show that the fringe width, as observed on the focal plane of the lens, will be $f\lambda/d$ where f is the focal length of the lens.

14.15 In Problem 14.14, show that if the plane (containing the holes) lies in the front focal plane of the lens, then the interference pattern will consist of exactly parallel straight lines. However, if the plane does not lie on the front focal plane, the fringe pattern will be hyperbolae.

14.16 In the Young's double-hole experiment calculate I/I_{max} where I represents the intensity at a point where the path difference is $\lambda/5$.